

Complexity Analysis of the Dutch Merchant Problem

Abstract

This document presents a computational complexity analysis of the Dutch Merchant Problem, demonstrating its NP-hardness via a polynomial-time reduction from the Prize-Collecting Traveling Salesman Problem (PCTSP) and establishing its NP-Completeness.

1 Introduction

The Dutch Merchant Problem (DMP) is a combinatorial optimization problem that models the strategic route and transaction decisions of a maritime merchant under multiple constraints. This work formally demonstrates that its associated decision problem is **NP-Complete**. The proof is structured in two main parts: (1) establishing its **NP-hardness** via a polynomial-time reduction from the Prize-Collecting TSP, and (2) proving its **membership in the class NP** by describing a polynomial-time verification algorithm. This fundamental result implies that no efficient exact algorithm (running in polynomial time) exists for solving arbitrary-sized instances unless $P = NP$, thus justifying the subsequent development of approximation algorithms and heuristics in the project's later phases.

2 Proof of NP-Hardness

2.1 Formal Definition of the Dutch Merchant Problem (DMP)

An instance of the DUTCHMERCHANT problem is defined by the tuple $(G, r, \mathcal{M}, \mathcal{P}, B, T_{max}, f)$ where:

- $G = (V, E)$ is a complete, undirected graph representing the port network. $V = \{v_0, v_1, \dots, v_n\}$, with v_0 being the origin port (Amsterdam).
- $r \in \mathbb{R}^+$ is the initial capital.
- $\mathcal{M}$ is the set of k commodity types.

- $\mathcal{P} = \{(c_{v,m}^{buy}, c_{v,m}^{sell}) \mid v \in V \setminus \{v_0\}, m \in \mathcal{M}\}$ defines the purchase and sale price for each commodity at each port.
- $B \in \mathbb{N}$ is the maximum cargo capacity of the ship (in commodity units).
- $T_{max} \in \mathbb{R}^+$ is the total maximum allowed duration for the expedition.
- $f : E \rightarrow \mathbb{R}^+$ assigns to each edge $e = (u, v)$ a travel time t_{uv} and an operational cost c_{uv} .

A **feasible solution** is a simple cycle $(v_0, v_{i_1}, \dots, v_{i_l}, v_0)$ and a sequence of buy/sell decisions at each visited port, such that: 1) The capacity B is never exceeded. 2) The capital never becomes negative after covering operational costs. 3) The total time (travel + operations) $\leq T_{max}$.

The **objective** is to maximize the final capital upon returning to v_0 .

2.2 Definition of the Reduced Problem: Restricted-DMP

For the reduction, we define a heavily restricted version of DUTCHMERCHANT, called RESTRICTED-DMP. This instance is obtained by fixing or eliminating the following parameters and decisions from the original problem:

1. **Commodities:** We fix $|\mathcal{M}| = 1$. Only a single generic commodity exists.
2. **Prices and Transactions:** We fix $c_v^{buy} = 0$ and $c_v^{sell} = p_v \geq 0$ for every port $v \in V \setminus \{v_0\}$. This implies that upon visiting v , a fixed net profit p_v is obtained automatically, with no intermediate purchase decision or capacity consumption.
3. **Capacity:** We fix $B = \infty$. Capacity ceases to be a constraint.
4. **Financial Constraint:** The intermediate non-negative capital condition is removed. The capital is simply $r + \sum p_v - \sum c_{uv}$.
5. **Time:** We fix $T_{max} = \infty$ or to a value sufficiently large to not restrict any possible tour. The edge costs c_{uv} represent capital *penalties* for travel.

Therefore, an instance of RESTRICTED-DMP is defined simply by $(G, v_0, \{p_v\}_{v \in V}, \{c_e\}_{e \in E})$, and its objective is to find a cycle starting and ending at v_0 , visiting a subset $S \subseteq V \setminus \{v_0\}$, that maximizes $\sum_{v \in S} p_v - \sum_{e \in \text{tour}} c_e$.

2.3 Polynomial-Time Reduction from Prize-Collecting TSP

Definition of PCTSP

An instance of the PRIZECOLLECTINGTSP (PCTSP) problem is given by $(H, s, \{\pi_u\}_{u \in U}, \{\omega_{ij}\}_{(i,j) \in F})$, where $H = (U, F)$ is a complete graph, $s \in U$ is the root node, $\pi_u \geq 0$ is the prize for visiting node u , and $\omega_{ij} \geq 0$ is the cost of traversing edge (i, j) . The goal is to find a cycle that starts and ends at s , visits a subset of nodes, and maximizes the sum of collected prizes minus the sum of traversal costs.

Theorem 1 (NP-Hardness of PCTSP): PRIZECOLLECTINGTSP is NP-hard. This follows directly from it being a generalization of the classic TRAVELINGSALESMANPROBLEM (TSP, NP-hard) where the prizes for all nodes except s are greater than the maximum possible sum of travel costs. Furthermore, the NP-hardness of this problem is well-established in the literature, as documented in studies on prize-collecting routing problems [1].

Construction of the Reduction

Given an arbitrary instance $I_P = (H, s, \{\pi_u\}, \{\omega_{ij}\})$ of PCTSP, we construct an instance $I_C = (G, v_0, \{p_v\}, \{c_e\})$ of RESTRICTED-DMP in polynomial time as follows:

1. For each node $u \in U$ in H , we create a corresponding port $v \in V$ in G . The root node s is mapped to the origin port v_0 .
2. The graph G is complete. For each edge (i, j) in H with cost ω_{ij} , we define the travel/operational cost in G as $c_{v_i v_j} = \omega_{ij}$.
3. For each port v corresponding to a node $u \neq s$, we define the fixed profit $p_v = \pi_u$.

This construction runs in $O(|U|^2)$ time.

Solution Equivalence and Optimality

There exists a bijective correspondence between feasible tours of I_P and I_C that preserves the objective function. Given a tour $T_P = (s, u_{i_1}, \dots, u_{i_m}, s)$ for I_P with value $\sum \pi_{u_{i_k}} - \sum \omega_{tour}$, the corresponding tour $T_C = (v_0, v_{i_1}, \dots, v_{i_m}, v_0)$ in I_C yields a final capital of $r + \sum p_{v_{i_k}} - \sum c_{tour}$. Since r is constant, maximizing final capital in I_C is equivalent to maximizing $\sum p_{v_{i_k}} - \sum c_{tour}$, which is identical to the objective of I_P . The inverse argument is analogous.

Lemma 1: There exists a tour in I_P with value $\geq K$ if and only if there exists a tour in I_C with final capital $\geq r + K$.

Reduction Conclusion

Since PCTSP is NP-hard (Theorem 1) and we have constructed a polynomial-time, optimality-preserving reduction from PCTSP to RESTRICTED-DMP, we conclude:

Theorem 2: RESTRICTED-DMP is NP-hard.

2.4 Complexity Preservation Upon Reintroducing Constraints

Theorem 2 demonstrates the NP-hardness of a simplified version (Π_{restr}) of our original problem (Π_{orig}). To definitively conclude that Π_{orig} is NP-hard, we must analyze why reintroducing each eliminated constraint does not transform

the problem into an easier one (in P). We demonstrate that, for each constraint, Π_{restr} is a **special case** of Π_{orig} under a specific configuration; therefore, Π_{orig} is at least as hard.

Let $\Pi_{orig}(X, Y, Z, \dots)$ be the general problem with constraints $X, Y, Z, \dots$, and Π_{restr} the problem without them. If Π_{restr} is NP-hard, and there exists an assignment of values for $X, Y, Z, \dots$ such that $\Pi_{orig}(val_X, val_Y, val_Z, \dots) \equiv \Pi_{restr}$, then Π_{orig} is NP-hard.

1. **Multiple Commodities** ($|\mathcal{M}| \geq 1$). In Π_{restr} , $|\mathcal{M}| = 1$. Consider an instance of Π_{orig} with $|\mathcal{M}| = k > 1$ where: i) the purchase prices for commodities $2, \dots, k$ are higher than any possible profit, making them non-profitable; ii) the sale price of commodity 1 is identical to p_v ; and iii) capacity is sufficient to carry only commodity 1. Any optimal solution will ignore commodities $2, \dots, k$ and behave as in Π_{restr} . Thus, Π_{restr} is a special case of Π_{orig} with multiple commodities, which does not reduce complexity.
2. **Finite Cargo Capacity** ($B \in \mathbb{N}$). In Π_{restr} , $B = \infty$. Setting B to a sufficiently large value (e.g., $B \geq \sum_{v \in V} 1$) in Π_{orig} replicates the unlimited capacity condition, as the limit will never be reached with a single unit-profit commodity. Reducing B restricts the solution space, adding a dimension of difficulty equivalent to the Knapsack problem, which is NP-hard. Adding an NP-hard constraint cannot simplify a problem already proven NP-hard.
3. **Intermediate Financial Constraint (Non-Negative Capital)**. In Π_{restr} this constraint is omitted. In Π_{orig} , if the initial capital r is greater than the maximum possible sum of travel costs, the constraint never activates, replicating Π_{restr} . Reducing r introduces a dynamic financial state that must be verified at each port, increasing the complexity of the state space, not reducing it.
4. **Strict Time Limit** ($T_{max} \in \mathbb{R}^+$). In Π_{restr} , $T_{max} = \infty$. Setting T_{max} larger than the duration of the longest possible tour in Π_{orig} renders the constraint inactive, replicating Π_{restr} . Imposing a strict time limit transforms the problem into a variant of the "Orienteering Problem" (also NP-hard), adding an additional layer of difficulty.
5. **Complex Buy/Sell Decisions**. In Π_{restr} , profit is automatic ($c_v^{buy} = 0, c_v^{sell} = p_v$). In Π_{orig} , fixing prices identically and defining that profit is obtained upon port arrival (without capacity consumption) simulates Π_{restr} . Allowing variable prices and strategic decisions introduces an arbitrage and inventory management problem, which is a strict and more difficult generalization.

Analysis Conclusion: For each constraint R_i , there exists a configuration of Π_{orig} that makes it equivalent to Π_{restr} . Any adjustment that genuinely activates R_i (e.g., reducing B , reducing T_{max}) **restricts** the feasible solution

space and/or adds an NP-hard optimization dimension. Therefore, Π_{orig} is no simpler than Π_{restr} ; it is a generalization that preserves and potentially increases complexity.

Theorem 3 (NP-Hardness of the General Problem): The DUTCH-MERCHANT problem (Π_{orig}) is NP-hard. **Proof:** Direct from Theorem 2 and the above analysis, which establishes that Π_{restr} is a special case of Π_{orig} . $\square$

2.5 Proof of Membership in NP

To establish that DUTCHMERCHANT is NP-Complete, in addition to being NP-hard (Theorem 3), we must prove it belongs to the class NP. That is, given a proposed solution to the problem, we can verify its feasibility and compute its objective value (the final capital) in **polynomial time** relative to the input size.

Certificate and Verifier

A certificate C for an instance of Π_{orig} must include:

1. A sequence of ports (a tour) $T = (v_0, v_{i_1}, v_{i_2}, \dots, v_{i_m}, v_0)$.
2. For each visited port v_{i_j} , a list of buy and sell operations per commodity type.

The size of this certificate is polynomially bounded: the tour has at most $|V|$ nodes, and operations at each port are for $|\mathcal{M}|$ commodities.

A verifier algorithm $V(I, C)$ must check the following points in polynomial time:

1. **Tour Validity:** Verify T is a cycle starting and ending at v_0 and that no internal port is repeated. This takes $O(|V|)$.
2. **Capacity Compliance:** For each leg between ports, calculate the total onboard load by summing purchases and subtracting sales. Verify it never exceeds B . This requires traversing the operation sequence once: $O(m \cdot |\mathcal{M}|)$.
3. **Financial Constraint Compliance:** Simulate the cash flow. Starting from r , for each port: subtract travel costs, add sale profits, subtract purchase costs, and verify non-negative capital. Also $O(m \cdot |\mathcal{M}|)$.
4. **Time Limit Compliance:** Sum travel times t_{uv} for all edges in T and port operation times. Verify the total does not exceed T_{max} . $O(m)$.
5. **Objective Calculation:** The certificate's value is the final capital after the last operation, obtained as the output of step 3.

Each operation is an addition, subtraction, or comparison executed a number of times proportional to the certificate size. Therefore, total verification time is polynomial in the input size I (which includes $|V|$, $|E|$, $|\mathcal{M}|$).

Lemma 3 (Membership in NP): There exists a verifier $V(I, C)$ that decides in polynomial time whether a certificate C is a feasible solution for Π_{orig} and what its value is. Therefore, $\Pi_{orig} \in \text{NP}$.

NP-Completeness Conclusion

Combining the NP-hardness result (Theorem 3) with NP membership (Lemma 3), we establish the final classification.

Theorem 4 (NP-Completeness): The decision problem associated with DUTCHMERCHANT is NP-Complete. **Proof:** The decision problem asks: "Does there exist a route and transaction plan with final capital $\geq K$?" By Lemma 3, this problem is in NP. By Theorem 3, it is NP-hard. Consequently, it is NP-Complete. $\square$

This classification formally justifies the use of non-exact algorithmic techniques (heuristics, approximations) in the project's subsequent phases.

3 Conclusions

This work presents a formal analysis of the computational complexity of the Dutch Merchant Problem. The main results are:

- A restricted version of the problem (RESTRICTED-DMP) was defined and proven NP-hard via a polynomial-time reduction from the Prize-Collecting TSP (Theorem 2).
- It was rigorously argued, constraint by constraint, that the NP-hard complexity is preserved when reintroducing all features of the general problem (Theorem 3).
- It was demonstrated that the associated decision problem belongs to the class NP, by presenting a polynomial-time verifiable certificate (Lemma 3 and Theorem 4).

Therefore, we conclude that the **Dutch Merchant decision problem is NP-Complete**. This finding provides the theoretical justification for the project's practical approach, which in the following phases will focus on the design, implementation, and empirical evaluation of approximation algorithms and heuristics to tackle realistic problem instances.

References

- [1] Alberto Marchetti-Spaccamela, Vincenzo Bonifaci, Stefano Leonardi, and Giorgio Ausiello. Prize-collecting traveling salesman and related problems. In *Handbook of Approximation Algorithms and Metaheuristics*, 2007.